

University of Niagara Falls Canada
Master of Data Analytics
SQL Databases
Assignment 2

The following questions evaluate your ability to:

- LO1. Apply partitioning and aggregation using windows functions.
- LO2. Design nested queries using SQL aggregate functions and windows functions.
- LO3. Employ functions that compare datapoints to their historical and future values.

Question 1

Write a query that calculates the total number of rentals for each film in the Sakila database and partitions these films into quartiles based on their rental counts. Your result must include the film title, the total number of rentals, and the quartile number (1, 2, 3, or 4) for each film.

Question 2

For each film category, write a query to identify the films with rental rates that exceed the average rental rate for that category. The output should display the film category name, film title, the film's rental rate, and the average rental rate computed for the category.

Question 3

Create a query that ranks films within each language by their replacement cost in descending order. Your result should include the film title, the language, the replacement cost, and the rank of each film within its language group.

Question 4

Develop a query to compute the interval, in days, between consecutive rentals for each customer in the Sakila database. The output should look like the followings:

Question 5

Write a query to determine the top ten film categories by total rental count. Your output should include the film category name and the total number of rentals for that category.

Rubric of assignment 2 - CPSC500 SQL Databases

Criteria	Excellent (10 points)	Good (8 points)	Satisfactory (6 points)	Fail (5 points or less)
Use of software tools (MySQL)	Effectively and comprehensively utilizes MySQL tools to complete the tasks.	Good use of MySQL tools with minor inefficiencies.	Basic use of required tools but may lack effectiveness or completeness.	Ineffective or incorrect use of MySQL tools.
Partitioning & aggregation	Successfully partitions customers into quartiles using correct SQL window functions.	Correctly partitions most customers, but with minor errors or inefficiencies.	Partitions customers, but with significant mistakes or missing logic.	Fails to properly partition customers or does not retrieve the required data.
Nested queries	Accurately retrieves product prices greater than the average, including correct use of nested queries and SQL functions.	Mostly correct, but contains minor logic or efficiency errors.	Retrieves some correct data but contains syntax or logical mistakes.	Query is incorrect, missing key logic, or fails to retrieve required data.
Ranking query	Assigns ranks correctly using SQL ranking functions with optimized query structure.	Assigns ranks correctly but may have minor inefficiencies.	Attempts ranking but with errors in ranking logic or syntax.	Ranking query is incorrect or does not return valid results.
Handling database features over time	Accurately retrieve the values of features using appropriate window functions.	Query is mostly correct but contains minor inefficiencies.	Attempts to retrieve future values but has logical or syntax errors.	Fails to retrieve the future values correctly.
Following SQL coding standards	All queries comply with SQL coding standards	Most of the queries meet the coding standards	Fair consideration of coding standards	Queries do not meet the SQL coding standards